

Evolution of respiratory function in Duchenne muscular dystrophy from childhood to adulthood.

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In Duchenne muscular dystrophy (DMD), it is still to be determined if specific timepoints can be identified during the natural evolution of respiratory dysfunction from childhood to adulthood and if scoliosis, steroid therapy and nocturnal noninvasive mechanical ventilation (NIMV) have any effect on it. In a 7-year retrospective study performed on 115 DMD patients (6-24 years), evaluated once or twice per year, with 574 visits in total, evolution mean curves of spirometry, lung volumes, spontaneous breathing and thoraco-abdominal pattern (measured by optoelectronic plethysmography) parameters were obtained by nonlinear regression model analysis. While predicted values of forced vital capacity, forced expiratory volume in 1 s, and peak expiratory flow decline continuously since childhood, during spontaneous breathing the following parameters become significantly different than normal in sequence: abdominal contribution to tidal volume (lower after 14.8 years), tidal volume (lower after 17.2 years), minute ventilation (lower after 18.1 years) and respiratory rate (higher after 22.1 years). Restrictive lung pattern and diaphragmatic impairment are exacerbated by scoliosis severity, slowed by steroids treatment and significantly affected by NIMV. Spirometry, lung volumes, breathing pattern and thoraco-abdominal contributions show different evolution curves over time. Specific timepoints of respiratory impairment are identified during disease progression. These should be considered when defining outcome measures in clinical trials and treatment strategies in DMD.

Spirometry correlates with physical activity in patients with Duchenne muscular dystrophy.

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Duchenne muscular dystrophy (DMD) is associated with progressive muscle weakness and respiratory decline. To date, studies have focused on respiratory decline and loss of ambulation as a metric of loss of skeletal muscle strength. However, new functional measures can assess skeletal muscle disease regardless of ambulatory status. The relationship between these tests and concurrent lung function is currently unexplored. To assess the correlation between spirometry measurements and functional muscle assessments such as accelerometry and quantitative muscle testing (QMT). Enrolled patients with DMD underwent accelerometry and QMT at study clinic visits. Any pulmonary function testing within 6 months of visit was obtained from the electronic medical record. The Spearman correlation coefficient was used to assess the relationship between spirometry and functional muscle testing. Forced vital capacity (FVC) and forced expiratory volume in 1 s (FEV1) demonstrated the strongest correlation with accelerometry. Both FVC and FEV1 showed a similar relationship to accelerometry when activity was divided into intensity categories, with low intensity and moderate-to-vigorous activity categories showing the strongest correlation. Maximal expiratory pressure (MEP) and FVC showed the most robust correlations with total QMT ($p < 0.001$ and $p < 0.01$, respectively). Lung function, specifically FVC percent predicted and FEV1 %p, shows a good correlation with upper and lower extremity skeletal muscle functional testing such as accelerometry and QMT.

Clinical implications of maximal respiratory pressure determinations for individuals with

Duchenne muscular dystrophy.

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To analyze the relationship between disease progression, pulmonary volumes, respiratory muscle strength (maximum inspiratory [MIP] and expiratory [MEP] pressure), and arterial blood gases for patients with Duchenne muscular dystrophy (DMD). An inception cohort study of pulmonary volumes, MIPs, and MEPs, correlated with age and PaCO₂ levels and with each other using linear and nonlinear regression analyses. Outpatient clinic. Fifty-two consecutive DMD patients who presented for regular evaluations at a regional DMD center. Maximum expiratory pressures were 47.7% +/- 10.9% of normal in the 167- to 14-year-old patients and decreased linearly thereafter ($MEP\% = -2.7 \times \text{age} + 73.8$; $p < .001$). Declines in MEP also correlated linearly with expiratory reserve volume ($p < .001$) and inversely with residual volume ($p < .001$). By contrast, MIP was 66.3% +/- 19.0% in the 357- to 14-year-old patients and then declined to 30.2% +/- 19.5% after age 14. No linear relationships were found with age but declines did correlate linearly with inspiratory reserve volume ($p < .001$) and total lung capacity ($p < .001$). PaCO₂ elevations correlated best with decreases in MIP ($p < .0001$) and appeared when MIP was below 30cmH₂O. Lung volume changes in DMD patients correlate with respiratory muscle weakness, and although inspiratory muscle dysfunction plays a key role in the development of chronic ventilatory insufficiency, reductions in expiratory muscle strength are the first signs of dysfunction and lead to the first episodes of respiratory failure.

Characterization of pulmonary function in Duchenne Muscular Dystrophy.

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Decline in pulmonary function in Duchenne Muscular Dystrophy (DMD) contributes to significant morbidity and reduced longevity. Spirometry is a widely used and fairly easily performed technique to assess lung function, and in particular lung volume; however, the acceptability criteria from the American Thoracic Society (ATS) may be overly restrictive and inappropriate for patients with neuromuscular disease. We examined prospective spirometry data (Forced Vital Capacity [FVC] and peak expiratory flow [PEF]) from 60 DMD patients enrolled in a natural history cohort study (median age 10.3 years, range 5-24 years). Expiratory flow-volume curves were examined by a pulmonologist and the data were evaluated for acceptability using ATS criteria modified based on the capabilities of patients with neuromuscular disease. Data were then analyzed for change with age, ambulation status, and glucocorticoid use. At least one acceptable study was obtained in 44 subjects (73%), and 81 of the 131 studies (62%) were acceptable. The FVC and PEF showed similar relative changes in absolute values with increasing age, i.e., an increase through 10 years, relative stabilization from 10-18 years, and then a decrease at an older age. The percent predicted, FVC and PEF showed a near linear decline of approximately 5% points/year from ages 5 to 24. Surprisingly, no difference was observed in FVC or PEF by ambulation or steroid treatment. Acceptable spirometry can be performed on DMD patients over a broad range of ages. Using modified ATS criteria, curated spirometry data, excluding technically unacceptable data, may provide a more reliable means of determining change in lung function over time.

Silencing XIST on the future active X: Searching human and bovine preimplantation embryos for the repressor.

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X inactivation is the means of equalizing the dosage of X chromosomal genes in male and female eutherian mammals, so that only one X is active in each cell. The XIST locus (in cis) on each additional X chromosome initiates the transcriptional silence of that chromosome, making it an inactive X. How the active X in both males and females is protected from inactivation by its own XIST locus is not well understood in any mammal. Previous studies of autosomal duplications suggest that gene(s) on the short arm of human chromosome 19 repress XIST on the active X. Here, we examine the time of transcription of some candidate genes in preimplantation embryos using single-cell RNA sequencing data from human embryos and qRT-PCR from bovine embryos. The candidate genes assayed are those transcribed from 19p13.3-13.2, which are widely expressed and can remodel chromatin. Our results confirm that XIST is expressed at low levels from the future active X in embryos of both sexes; they also show that the XIST locus is repressed in both sexes when pluripotency factors are being upregulated, during the 4-8 cell and morula stages in human and bovine embryos - well before the early blastocyst (E5) when XIST on the inactive X in females starts to be upregulated. Our data suggest a role for DNMT1, UHRF1, SAFB and SAFB2 in XIST repression; they also exclude XACT and other 19p candidate genes and provide the transcriptional timing for some genes not previously assayed in human or bovine preimplantation embryos.

Xist noncoding RNA could act as a tumor suppressor gene in patients with classical Hodgkin's disease.

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Xist is a long noncoding RNA involved in the X chromosome inactivation in females. It may act as an onco-suppressor gene in hematologic malignancies, and its activity is strongly dependent from SATB1 gene expression. However, its potential role in Hodgkin's disease (HD) onset and progression is unknown. Three gene expression microarray datasets were analyzed for the expression of Xist and SATB1 in patients with classical HD, namely, GDS4222 (130 patients and 54,000 gene features), GSE39134 (29 patients and 54,000 features), and E-MEXP-507 (29 patients and 27,648 probes). The first two were oligonucleotide arrays (platform: Affymetrix gene chip HG-U133-Plus2), whereas the latter was a cDNA two-channel array (platform: OncoChip. v2). Summary and time-dependent receiver operating characteristic (ROC) analysis were applied to obtain a summary measure (summary area under the ROC curve [sAUC]) of association between gene expression and unfavorable patient outcome in each probe set. Xist was overexpressed among females in each data set. A slight overexpression was associated with a good prognosis both in males (sAUC = 0.75, 95% confidence interval [CI]: 0.70-0.80) and at a lesser extent, in females (sAUC = 0.64, 95% CI: 0.59-0.69). However, this finding was limited to the analysis of the biggest database (GDS4222). No association was found between Xist and SATB1 expression. A reactivation of Xist might act as an onco-suppressor gene in male patients with HD, which seems independent from SATB1 expression. The possibility that Xist could contribute to the better survival of female patients should also be investigated.

Emerging similarities in epigenetic gene silencing by long noncoding RNAs.

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Long noncoding RNAs (lncRNAs) such as Xist, Air, and Kcnq1ot1 are required for epigenetic silencing of multiple genes in cis within large chromosomal domains, including distant genes located hundreds of kilobase pairs away. Recent evidence suggests that all three of these lncRNAs are functional and that they silence gene expression, in part, through an intimate interaction with chromatin. Here we provide an overview of lncRNA-dependent gene silencing, focusing on recent findings for the Air and Kcnq1ot1 lncRNAs. We review molecular evidence indicating that these lncRNAs interact with chromatin and correlate their presence with specific histone modifications associated with gene silencing. A general model for a lncRNA-dependent gene-silencing mechanism is presented based on the apparent ability of lncRNAs to recruit histone-modifying activities to chromatin. However, alternate mechanisms may be required to explain the silencing of some lncRNA-dependent genes. Finally, we discuss unanswered questions and future perspectives associated with these enigmatic lncRNA molecules.

Characterization of a murine gene expressed from the inactive X chromosome.

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In mammals, equal dosage of gene products encoded by the X chromosome in male and female cells is achieved by X inactivation. Although X-chromosome inactivation represents the most extensive example known of long range cis gene regulation, the mechanism by which thousands of genes on only one of a pair of identical chromosomes are turned off is poorly understood. We have recently identified a human gene (XIST) exclusively expressed from the inactive X chromosome. Here we report the isolation and characterization of its murine homologue (Xist) which localizes to the mouse X inactivation centre region and is the first murine gene found to be expressed from the inactive X chromosome. Nucleotide sequence analysis indicates that Xist may be associated with a protein product. The similar map positions and expression patterns for Xist in mouse and man suggest that this gene may have a role in X inactivation.

RNA binding proteins implicated in Xist-mediated chromosome silencing.

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Chromosome silencing by Xist RNA occurs in two steps; localisation in cis within the nuclear matrix to form a domain that corresponds to the territory of the inactive X chromosome elect, and transduction of silencing signals from Xist RNA to the underlying chromatin. Key factors that mediate these processes have been identified in a series of recent studies that harnessed comprehensive proteomic or genetic screening strategies. In this review we discuss these findings in light of prior knowledge both of Xist-mediated silencing and known functions/properties of the novel factors.

Xist drives spatial compartmentalization of DNA and protein to orchestrate initiation and maintenance of X inactivation.

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X chromosome inactivation (XCI) is the process whereby one of the X chromosomes in female

mammalian cells is silenced to equalize X-linked gene expression with males. XCI depends on the long noncoding RNA Xist, which coats the inactive X chromosome in cis and triggers a cascade of events that ultimately lead to chromosome-wide transcriptional silencing that is stable for the lifetime of an organism. In recent years, the discovery of proteins that interact with Xist have led to new insights into how the initiation of XCI occurs. Nevertheless, there are still various unknowns about the mechanisms by which Xist orchestrates and maintains stable X-linked silencing. Here, we review recent work elucidating the role of Xist and its protein partners in mediating chromosome-wide transcriptional repression, as well as discuss a model by which Xist may compartmentalize proteins across the inactive X chromosome to enable both the initiation and maintenance of XCI.
